

# JAYLEEN JIANG

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## EDUCATION

### Mount Holyoke College

Bachelor of Arts in Mathematics

GPA: 3.99/4.00

Coursework: Probability; Optimization; Stochastic Processes; Measure Theory; Dynamical Systems; Numerical Analysis

South Hadley, MA

Expected May 2027

## RESEARCH INTERESTS

Applied Probability, Scientific Computing, Dynamical Systems, Optimization

## PUBLICATIONS

**Jiang, J.**, Li, Y. *Numerical Study of an Energy Cascade Model Derived from a Dispersive Equation*. (Ready for submission.)  
Sanders, L. D., Pittman-Polletta, B., **Jiang, J.** *Computational Modeling of Distal Speech Rate Effects in Speech Segmentation*.  
(Submitted to *Attention, Perception, & Psychophysics*.)

## RESEARCH EXPERIENCE

### Cross-Spectral Image Correspondence for Industrial Robot Perception

University of California, Los Angeles

Advisor: **Sponsored by Analog Devices**

Jun. 2026 – Aug. 2026

- Attended the RIPS 2026 program at IPAM; formulated visible-infrared image correspondence for factory robot perception.
- Developed an evaluation framework to benchmark classical and learned feature matching methods under cross-spectral failure modes such as thermal hotspots, low texture, specular surfaces, and repeated structures.
- Explored matching pipelines that combine descriptor confidence, geometric consistency, and abstention criteria to reduce false correspondences in safety-critical robotic vision tasks.

### Numerical Study of Energy Cascades in Nonlinear Schrödinger Systems

UMass, Amherst

Advisor: **Prof. Yao Li**

Feb. 2025 – Present

- Built a high-throughput SIMD Monte Carlo integrator for the stochastic NLS energy-cascade chain, enabling the first numerical study of long chains beyond the solved three-mode case.
- Characterized the long-chain nonequilibrium steady state: local-Gibbs equilibrium in the bulk and anomalous sub-diffusive heat transport (insulating, unlike FPU lattices).
- Developed a mesh-free neural-network Fokker–Planck solver for the three-mode system, validated against the exact Gibbs measure and used to extract the generator’s spectral gap and eigenfunction.

### Fairness-Constrained Optimization for EV Charger Placement

University of California, Berkeley

Advisor: **Prof. Ming Gu**

Jun. 2025 – Aug. 2025

- Formulated EV charger deployment as a multi-objective optimization problem on a weighted transportation network, balancing accessibility, utilization, and distributional fairness.
- Integrated demand forecasts derived from demographic and infrastructure data into the optimization framework to account for spatial heterogeneity in charging needs.
- Developed a scalable heuristic algorithm combining candidate-site reduction, greedy construction, and local improvement to generate high-quality solutions for large-scale instances.

### Computational Neural Mechanisms of Distal Speech Rate Effects

Mount Holyoke College

Advisor: **Prof. Ben Pittman-Polletta**

Sept. 2024 – Dec. 2025

- Analyzed 128-channel EEG/ERP data from 21 participants, showing distal speech rate modulates early neural responses to reduced function words (N1, N280).
- Developed a semi-Markov particle filter model incorporating rate-dependent duration priors to infer perceptual state transitions from continuous speech.
- Demonstrated convergence between computational predictions and ERP findings, providing a mechanistic account of speech-rate-dependent word segmentation.

## SELECTED PROJECTS

### Evolution of Costly Cooperative Behaviors in Biological Systems

Independent Study Project

- Formulated ODE models incorporating reproduction dynamics, cost-benefit tradeoffs, and kin recognition; derived continuous recognition coefficients to smooth invasion threshold behavior.
- Built NetLogo agent-based simulations with genotype tracking, energy budgets, and stochastic reproduction to validate analytical predictions.

### Stochastic Mobility-Based Compartmental Model

MATH-646 Course Project

- Constructed a stochastic continuous-time Markov chain extension of a mobility-based SIRS epidemic model and derived its law-of-large-numbers ODE limit and central-limit diffusion approximation.
- Showed that mobility heterogeneity acts as a regularizing mechanism—reducing endemic prevalence, suppressing equilibrium noise, and raising the extinction probability above the homogeneous  $1/R_0$  baseline.

## TEACHING EXPERIENCES

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|---------------------------------------------------------------------------------------------------|-----------------------|
| <b>Teaching Assistant</b> , Mount Holyoke College                                                 | Jul. 2024 – Present   |
| Courses: <i>Calculus I – III, Intermediate Statistics, Mathematical Statistics, Real Analysis</i> |                       |
| <b>Course Tutor</b> , Vision Academy                                                              | Jun. 2023 – Aug. 2023 |
| Courses: <i>A-Level Mathematics, A-Level Advanced Mathematics</i>                                 |                       |

## CONFERENCE & TALKS

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| <b>Hudson River Undergraduate Mathematics Conference 2026</b>                                   | Apr. 2026  |
| Talk title: <i>Monte Carlo Density Estimation for the Stochastic NLS Energy Cascade System.</i> |            |
| <b>Nebraska Conference for Undergraduate Wisdom in Mathematics 2026</b>                         | Feb. 2026  |
| Talk title: <i>Optimal Fair Deployment Strategy for Electric Vehicle Charging Stations.</i>     |            |
| <b>Women in Mathematics in New England 2025</b>                                                 | Sept. 2025 |
| Talk title: <i>Neural Dynamics of Word Segmentation.</i>                                        |            |

## HONORS

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|--------------------------------------------------------------------|------|
| • Sarah Williston Award (top 1% in class year), MHC                | 2026 |
| • Mildred L. Sanderson Prize for Excellence in Mathematics, MHC    | 2025 |
| • Jennifer Landry '93 Award for Compassion in Math Education, MHC  | 2025 |
| • Eleanor Graham Claus Scholarship, Vong-ling Lee Scholarship, MHC | 2024 |

## SKILLS

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**Programming:** Python, C++, R, MATLAB, NetLogo, PyTorch, OpenMP  
**Language:** Mandarin (native), English (fluent), French (conversational)